Job Market & Salary Intelligence

End-to-End Data Analysis & Machine Learning Project Report



1. Project Objective

This project analyzes a job-postings dataset to understand the current job market and salary landscape. It cleans and validates raw job-posting records, engineers useful features, performs exploratory data analysis, studies in-demand skills, and builds machine learning models (Linear Regression and Random Forest) to predict average salary from job attributes such as experience, role, location, skills, and employment type.

2. Dataset Overview

The dataset contains 1,000 job postings with the following raw columns: Job_ID, Job_Title, Company, Location, Salary_Min, Salary_Max, Experience, Skills, Employment_Type, Posted_Date. Cleaning steps addressed missing salaries/skills/locations, duplicate records, inverted Salary_Min/Salary_Max pairs, and extreme salary outliers (capped using the IQR method).

3. Methodology / Workflow

The pipeline follows 13 stages: data loading → inspection → cleaning → missing-value handling → validation → outlier handling (IQR capping) → feature engineering (Experience_Years, Avg_Salary, Posted_Month) → skill extraction → exploratory visualization → correlation analysis → machine learning (Linear Regression & Random Forest) → model evaluation → business summary and export. See the workflow diagram on the cover page.

4. Headline Business Metrics

Metric	Value
Total Job Postings	1,000
Average Salary	Rs. 11,55,227.92
Median Salary	Rs. 11,24,336.00
Minimum Average Salary	Rs. 3,64,512.50
Maximum Average Salary	Rs. 20,28,114.44
Most Common Job Title	Business Analyst
Most Common Location	Pune
Most In-Demand Skill	SQL

5. Data Quality: Missing Values & Outliers

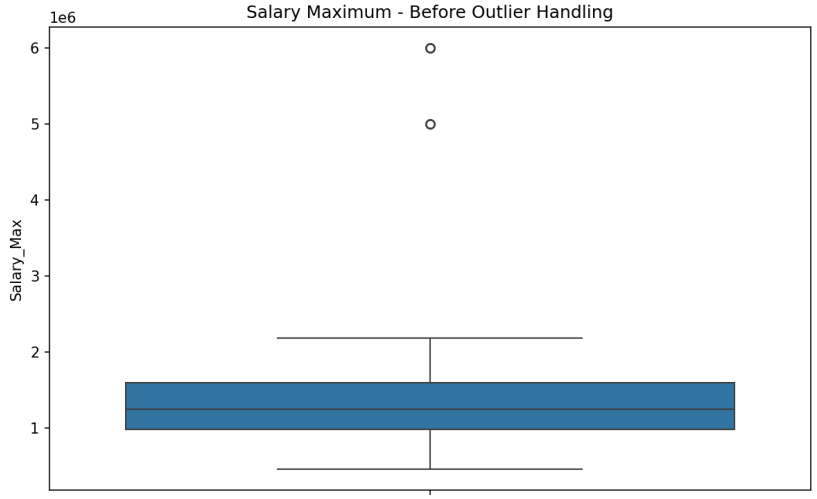
5.1 Missing Values (Before Cleaning)

Missing values were found in Location, Salary_Min, Salary_Max, and Skills. These were imputed using median (salary fields), mode (location), and a constant placeholder (skills).



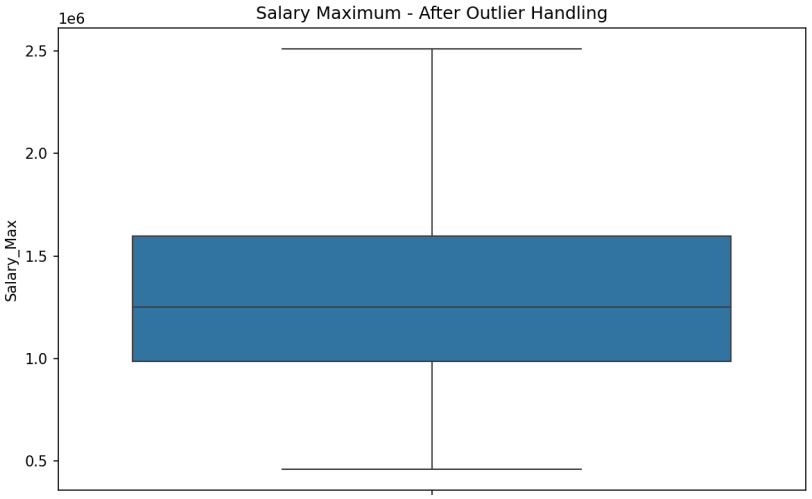
5.2 Salary Outliers (Before Capping)

A boxplot of Salary_Max reveals several extreme outlier postings far above the normal range.



5.3 Salary Outliers (After IQR Capping)

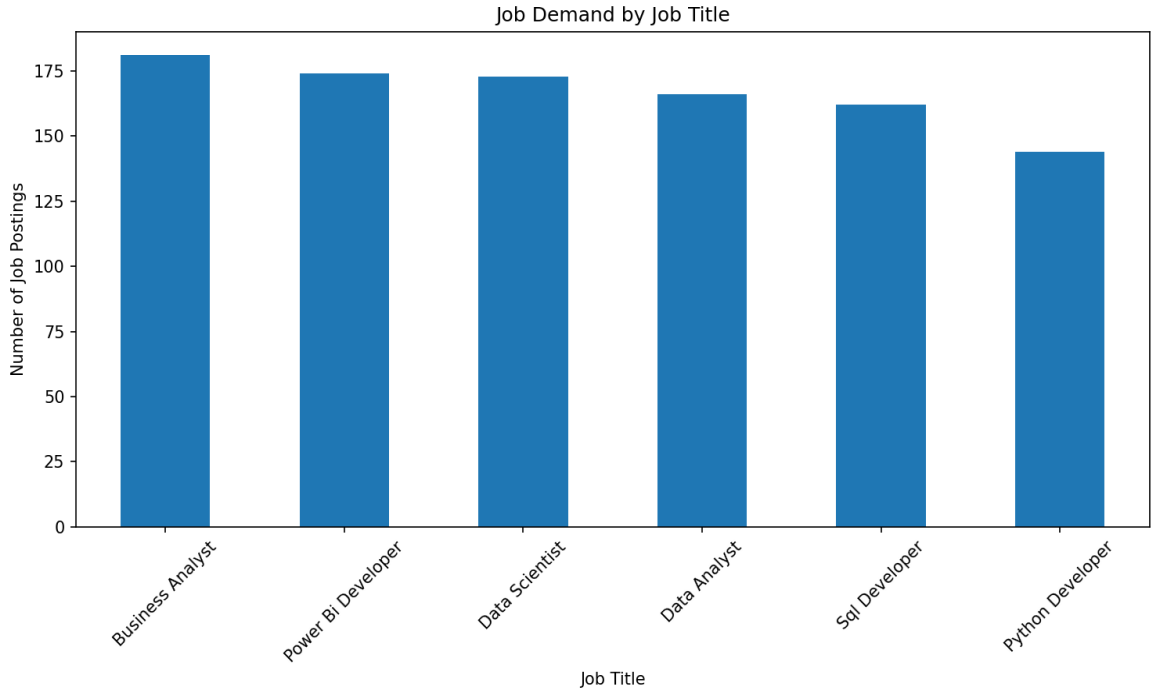
Outliers were capped to the IQR-based lower/upper bounds, producing a cleaner distribution without deleting any job records.


Salary Maximum - After Outlier Handling
1e6
2.5
2.0
1.5
1.0
0.5
Salary_Max

6. Exploratory Data Analysis

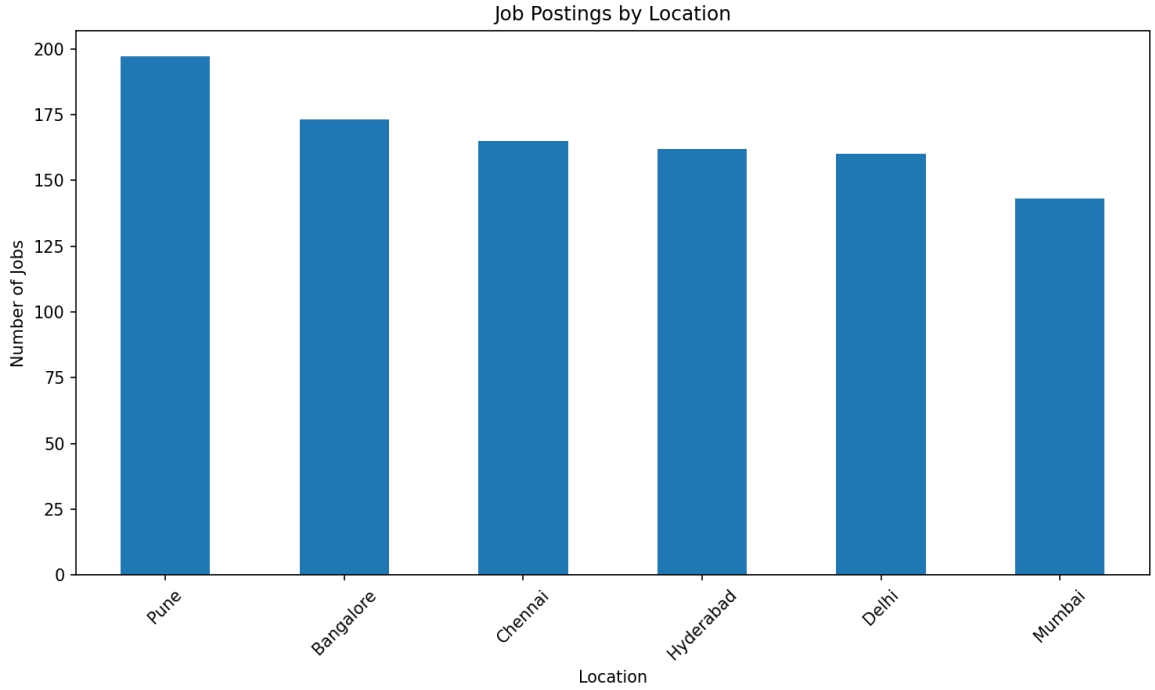
6.1 Job Demand by Job Title

Number of postings per role.



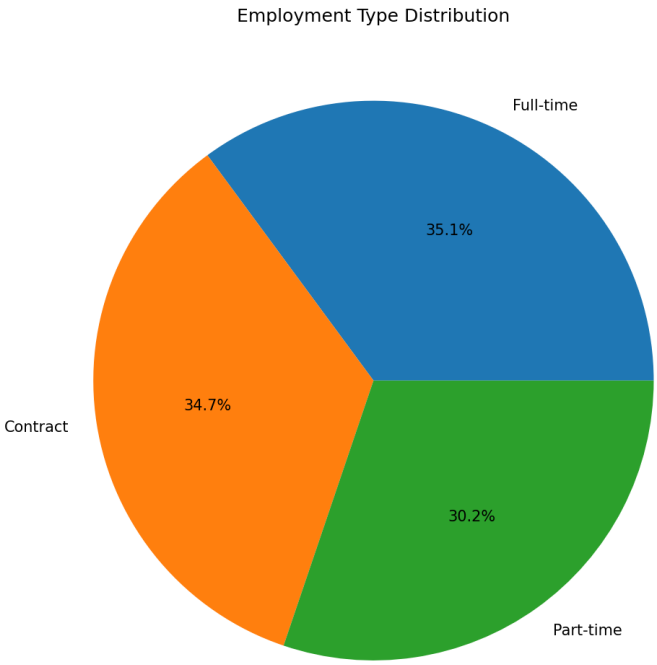
6.2 Job Postings by Location

Distribution of postings across cities.



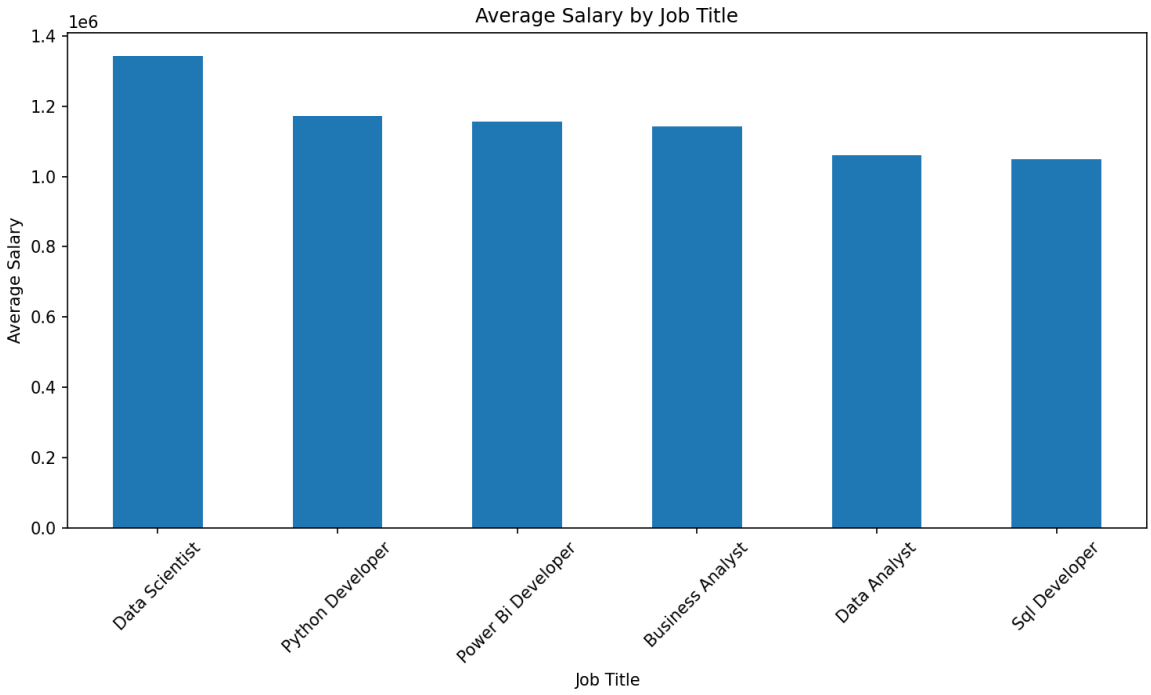
6.3 Employment Type Split

Share of Full-time, Contract and Part-time postings.

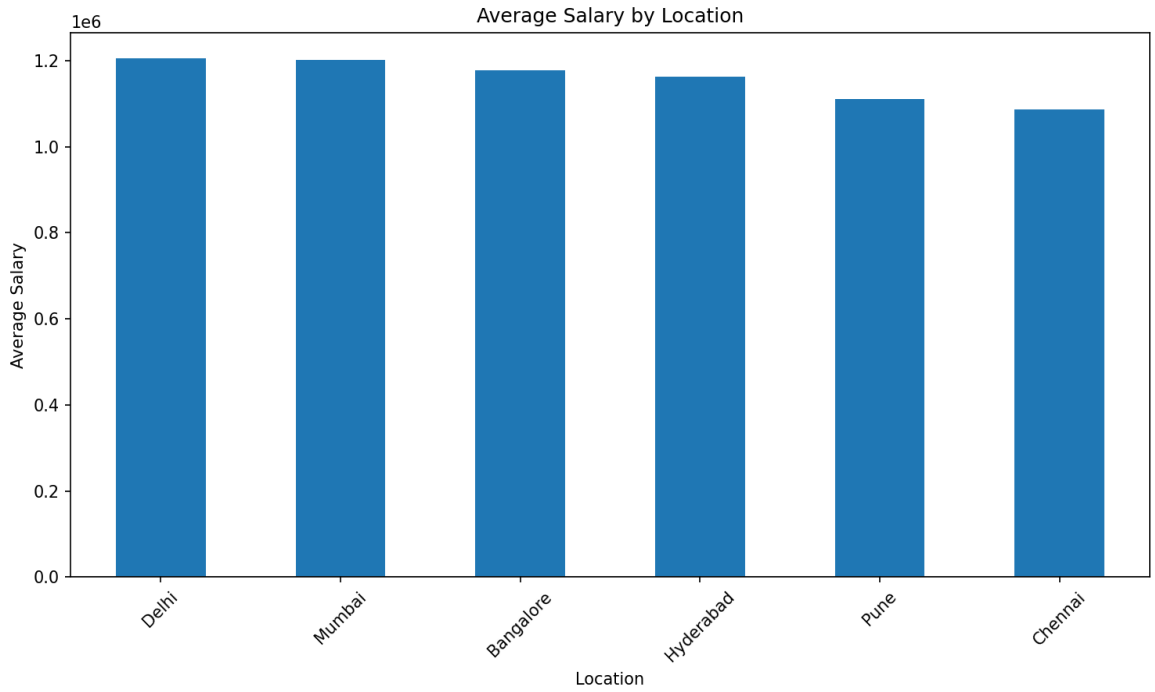


6.4 Average Salary by Job Title

Data Scientist commands the highest average salary among the roles analyzed.

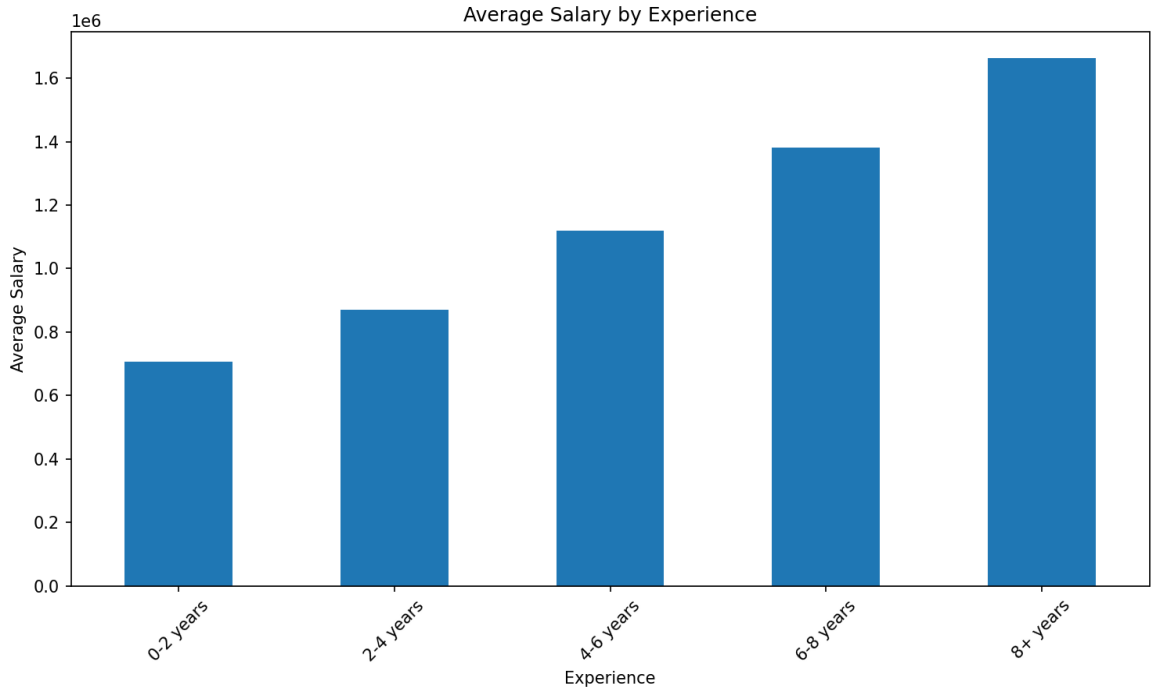


6.5 Average Salary by Location



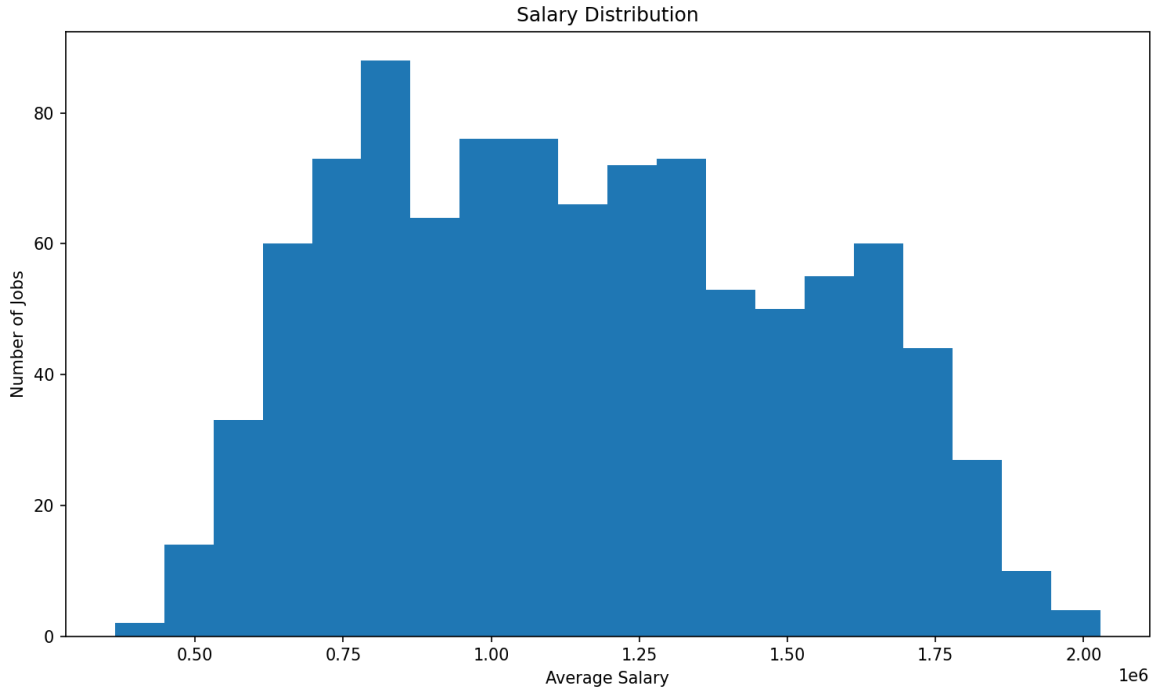
6.6 Average Salary by Experience Band

Salary rises consistently with experience band, as expected.



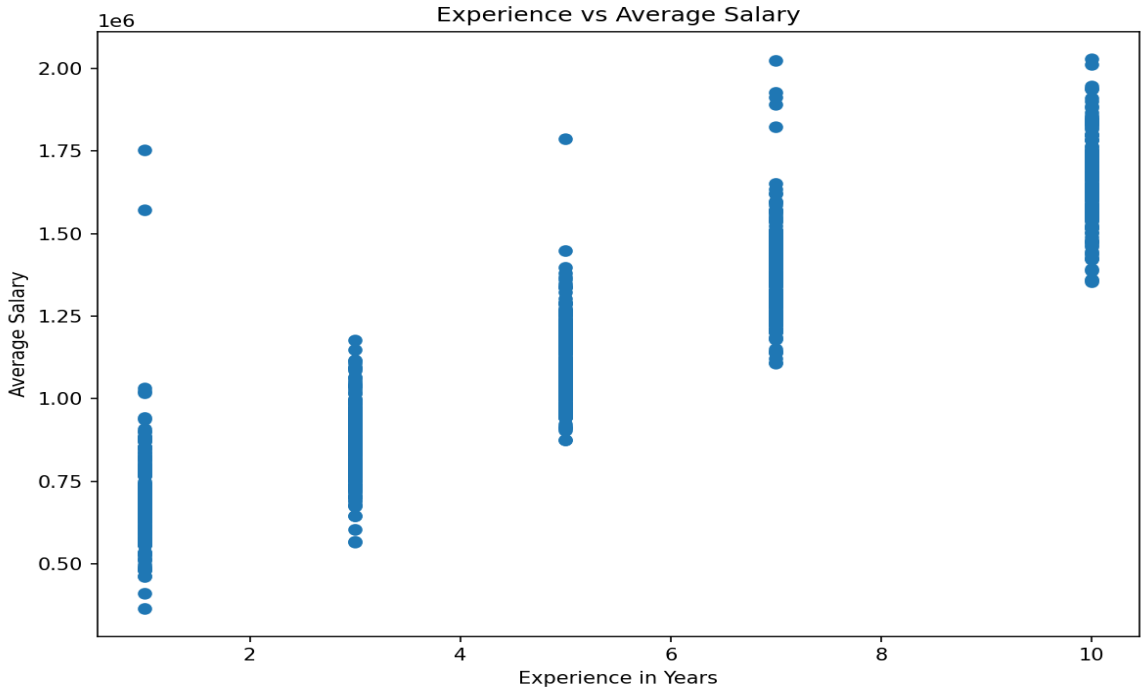
6.7 Salary Distribution

Histogram of average salary across all postings.



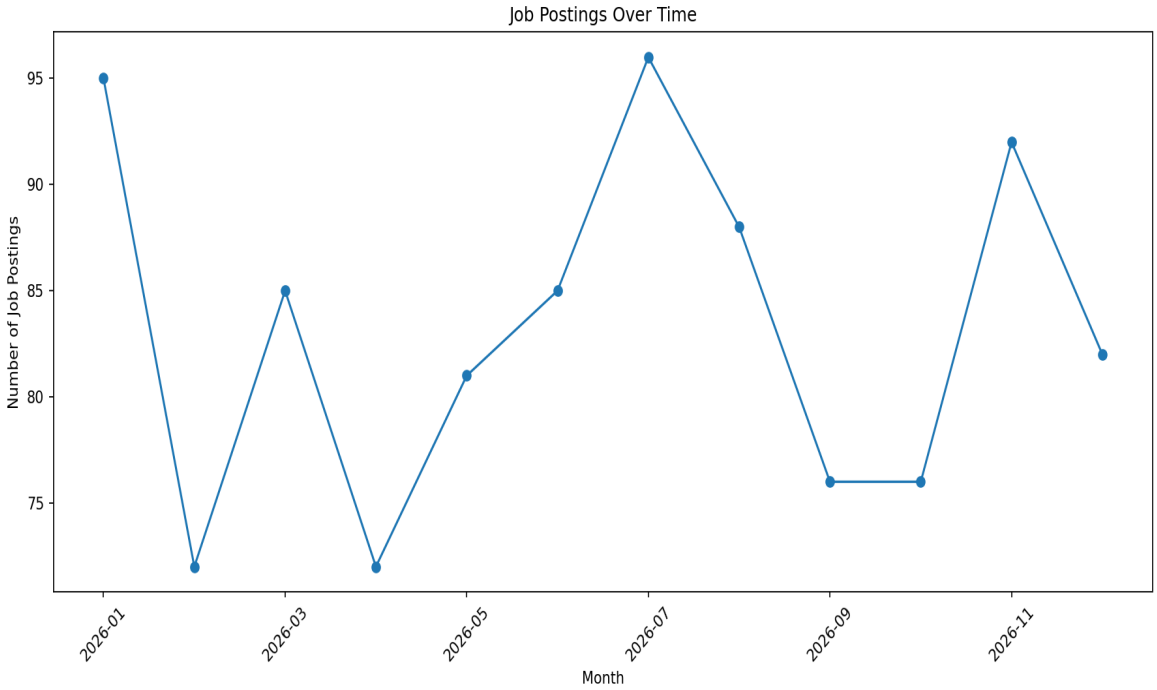
6.8 Experience vs Average Salary

Scatter plot showing a clear positive relationship, the basis for the regression model.



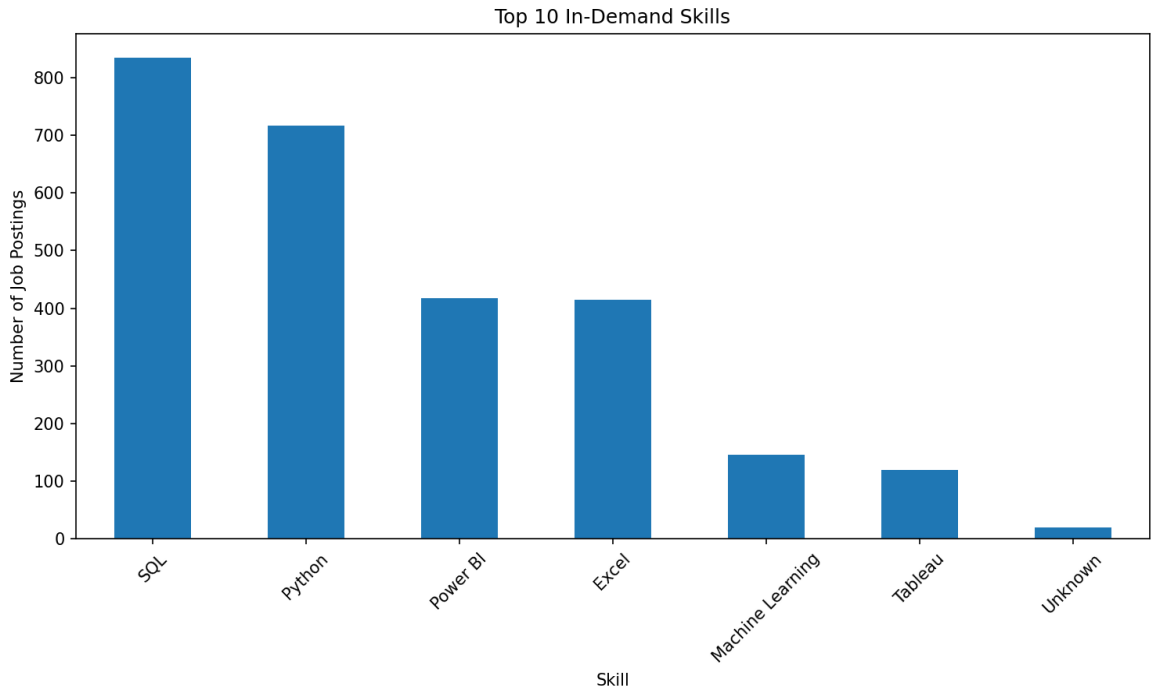
6.9 Job Postings Over Time

Monthly trend of postings across the year.



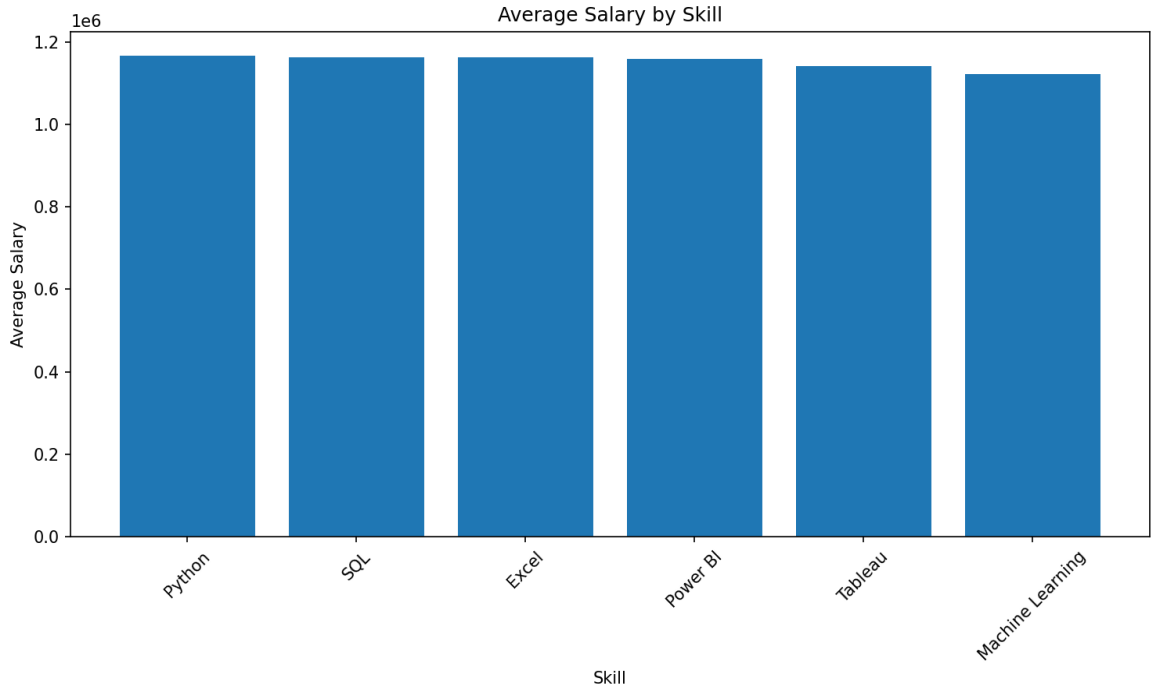
6.10 Top In-Demand Skills

SQL and Python dominate the skill requirements across postings.



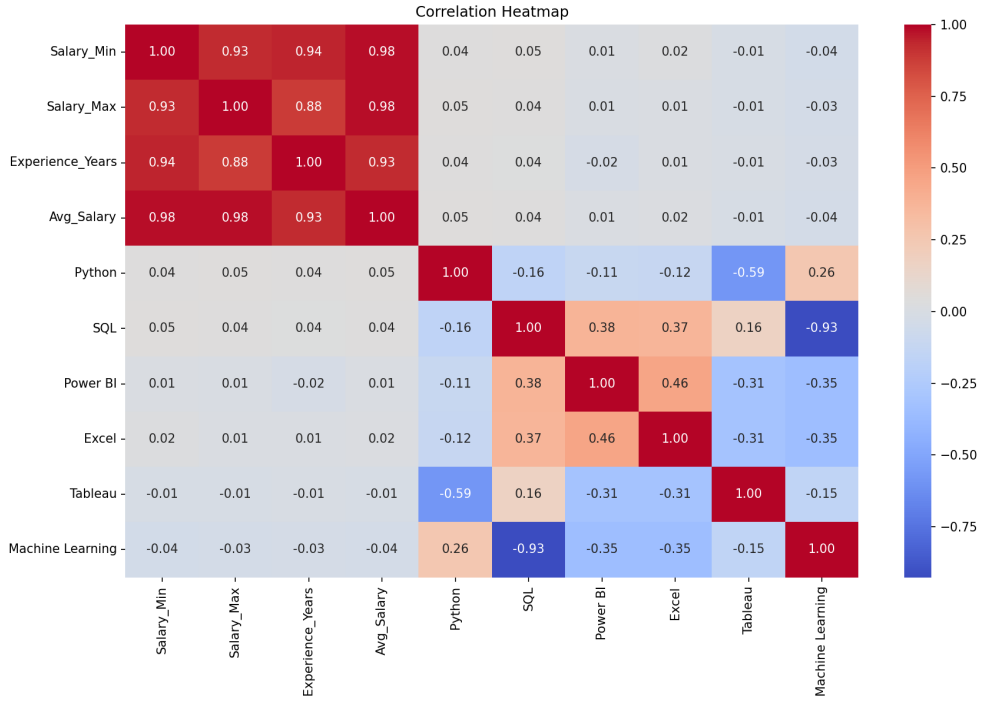
6.11 Average Salary by Skill

Machine Learning and Python postings correlate with higher average pay.



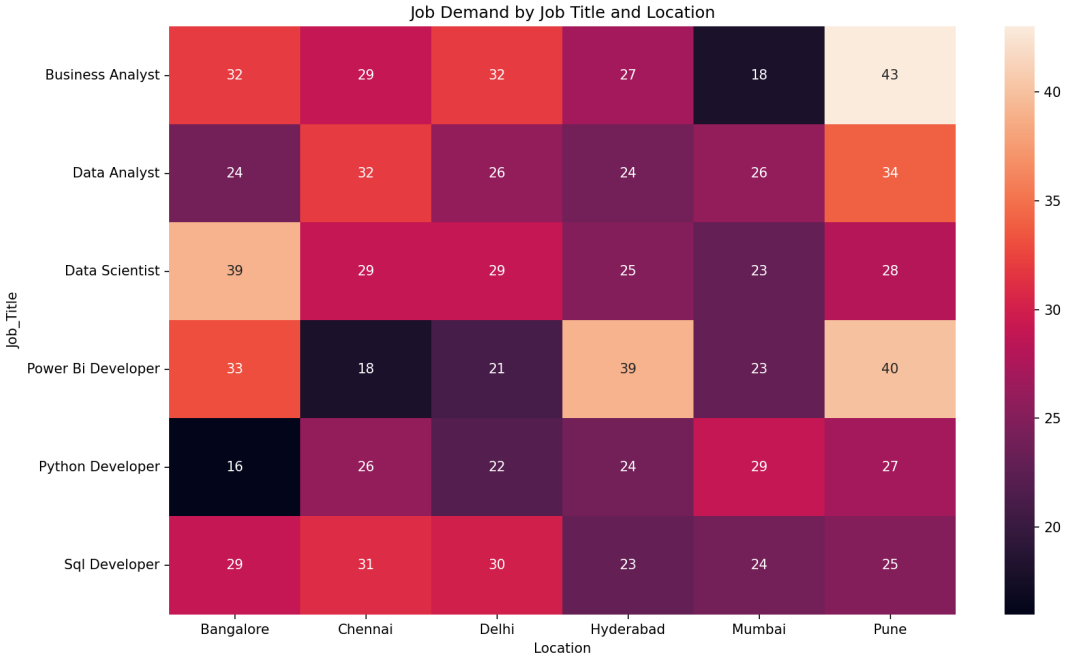
6.12 Correlation Heatmap

Correlation between numeric/salary and skill-flag columns.



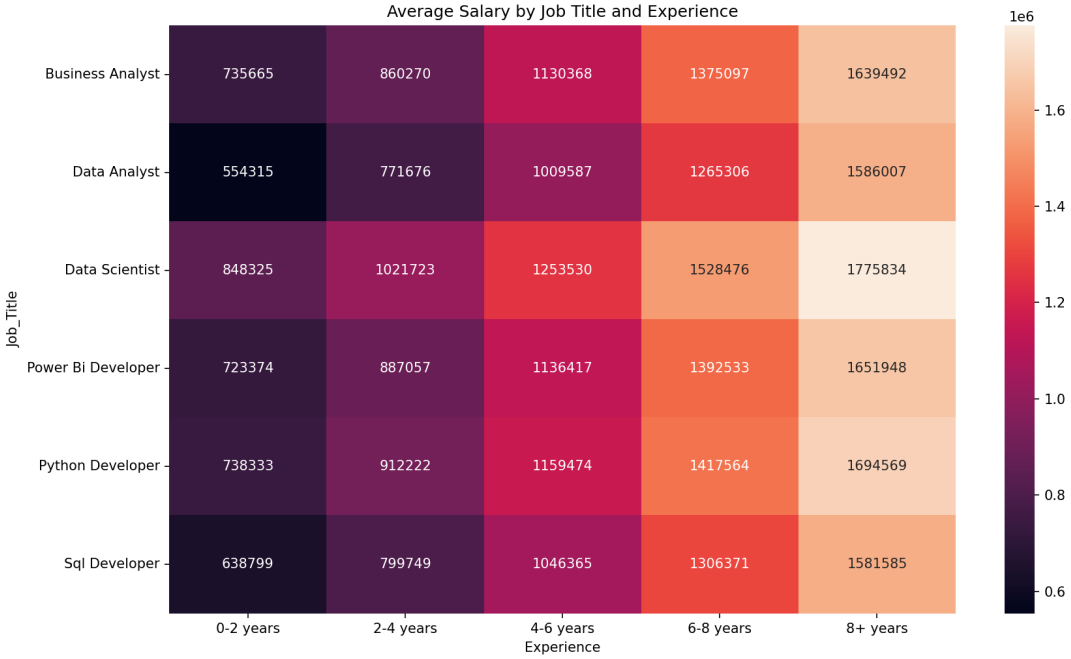
6.13 Job Title vs Location Heatmap

Cross-tabulation of postings by job title and location.



6.14 Salary by Job Title & Experience

Pivot table of average salary across job title and experience band.



7. Machine Learning: Salary Prediction

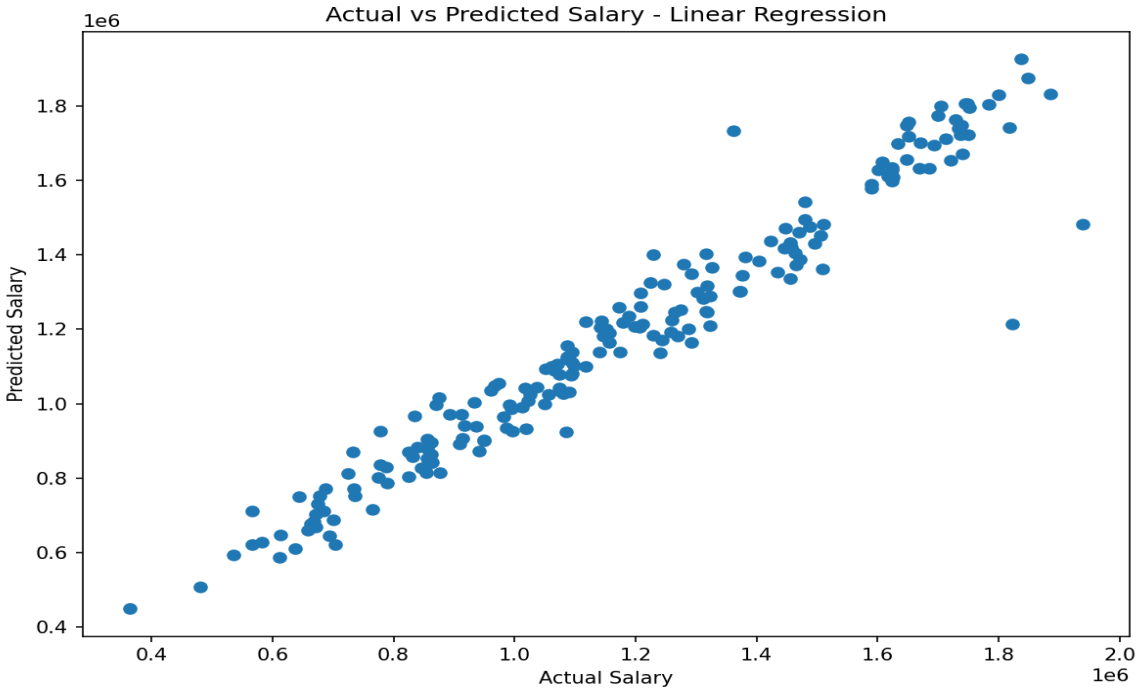
Target variable: **Avg_Salary**. Features used: Experience_Years, Job_Title, Location, Skills, Employment_Type — categorical fields one-hot encoded via a scikit-learn ColumnTransformer + Pipeline. Salary_Min and Salary_Max were **excluded** from the feature set since Avg_Salary is directly derived from them (this would otherwise cause target leakage). Data was split 80% train / 20% test (random_state=42).

Model	MAE	RMSE	R² Score
Linear Regression	53,682.56	84,154.08	0.9443
Random Forest	61,964.76	92,616.91	0.9326

Interpretation: Linear Regression slightly outperforms Random Forest on this dataset (higher R^2, lower error), suggesting salary is largely explained by additive, linear relationships between experience, role, location and skills, with limited non-linear interaction effects for Random Forest to exploit.

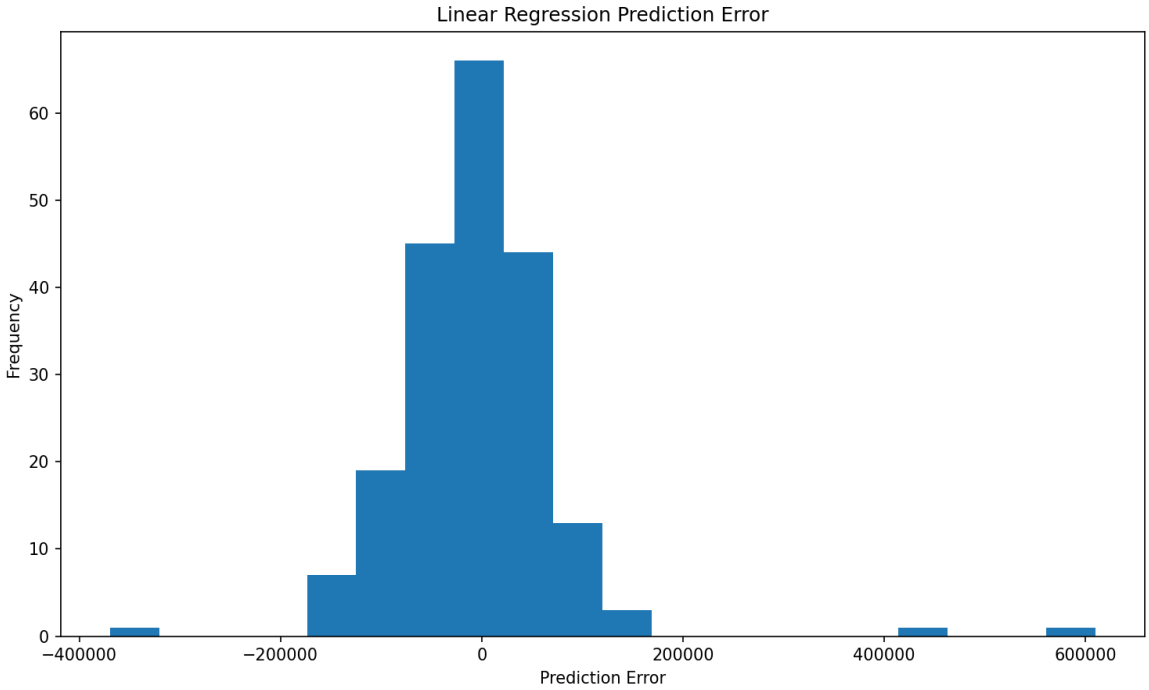
7.1 Actual vs Predicted Salary (Linear Regression)

Points close to the diagonal indicate accurate predictions.

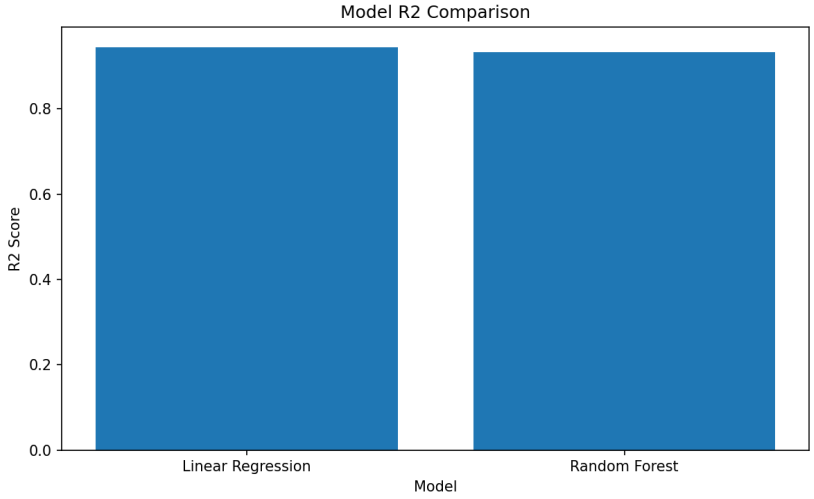


7.2 Prediction Error Distribution

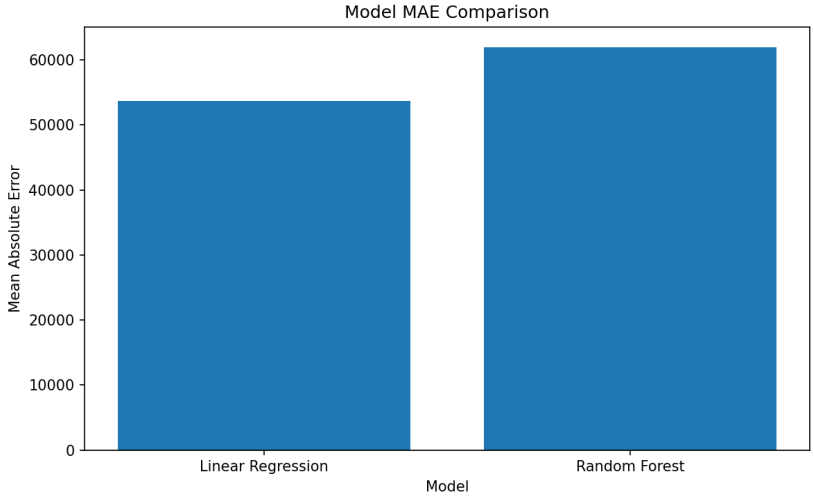
Errors are roughly centered around zero, indicating no major systematic bias.



7.3 Model R² Comparison



7.4 Model MAE Comparison



8. Skill Demand Summary

Skill	Job Postings
SQL	834
Python	716
Power BI	418
Excel	414
Machine Learning	146
Tableau	120
Unknown	20

9. Technologies & Libraries Used

Library	Purpose
pandas	Data loading, cleaning, transformation, export
numpy	Numerical operations, IQR bounds, outlier capping
matplotlib	Core charting (bar, line, scatter, histogram, pie)
seaborn	Boxplots and heatmaps (missing values, correlation, pivot)
scikit-learn	OneHotEncoder, Pipeline, Linear Regression, Random Forest, metrics
reportlab	This PDF report

Full details in Technologies_and_Libraries.md in the docs/ folder.

10. Key Insights & Conclusion

• Business Analyst and Pune lead in posting volume among roles and locations respectively.
• SQL and Python are the two most in-demand skills, appearing in the large majority of postings.
• Salary increases steadily with experience, confirming a strong positive relationship.
• Data Scientist and Python Developer roles command the highest average salaries.
• Linear Regression achieved $R^2 = 0.9443$, explaining over 94% of salary variance from experience, role, location, skills and employment type — a strong predictive result for this dataset.